

# Image Texture Recognition Based on Median Filtering and Hough Transform

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**Abstract**—This paper introduces a new computer recognition algorithm for image texture recognition. Using this algorithm, the texture characteristic of image is extracted exactly for analysis. 2-d bidirectional median filtering is proposed to make image de-noising an ideal state. An image correction algorithm with Hough transfer on local computation is presented to obtain an image with better position state. An approach based on local extreme value is proposed to search for feature points of image texture to further recognize each parameter of image texture. At last, this paper takes example for fabric density recognition, and gives the result and discusses detail process. The result shows that this approach is correct and efficient, and has extensive application.

**Keywords:** *Median Filtering; Hough transform; Image texture; Recognition; density.*

## I. INTRODUCTION

Image texture recognition is an important field of computer pattern recognition. Its research results have significance for quality detection and evaluating system. These researches fall into two general categories by now[1-2]. One is space domain. The image gray value is extracted for analysis and recognition. The representative approaches of space domain are geometric method[3], gray level co-occurrence matrix[4], neural network[5], probability statistics[6], and fuzzy clustering algorithm[7] and so on. The common characteristics of these algorithms is that the pixel gray value denoting image texture feature is extracted, and the texture feature is recognized by corresponding computation. Due to the pixel point gray value of image feature is extracted with this space domain approaches, the recognition results are more exact in theoretically. In fact, image appears much noise and skew disturbing by many factors. Also the accuracy of gray value is affected. Thus the recognition result will produce more errors.

The other is frequency domain analysis. The image is transformed to certain spectrum for analysis and recognition. At first, Fourier transform was main algorithm[8-9]. But this algorithm was so difficult to recognize dynamic window characteristic that this algorithm couldn't apply to practice. Later the wavelet algorithm[10-13] was used to recognize image texture, and once became research hotspot in frequency domain analysis. Though this algorithm solved the problem of dynamic window recognition, it needed massive calculations to perform real-time spectrum transform and still needed secondary recognition to analyses spectrum. The recognition speed with wavelet algorithm was not

ideal. In frequency domain analysis also appeared filtering algorithm[14-15]. The more representative approach was Gabor algorithm. Actually it is Fourier transform added windows. Though Gabor algorithm had the handling capacity with dynamic window, the problem of large calculation and secondary recognition was also exist. Otherwise, some scholars did many recognition researches on optical principle[16-17] depending on a great many sophisticated optical elements. This algorithm was simple, but the system was complex and cost was expensive.

In a word, a more perfect algorithm to recognize image texture is not proposed by now. Frequency domain analysis has disadvantages of large calculation and secondary recognition, and ultimately the image texture is difficult to be recognized rapidly and exactly. Space domain analysis is simple and extracts image features only once, and it is the main research direction for image texture recognition. But also has three main problems. One is image enhancement processing is not efficient, so that the definition is not enough to continue recognition. Two is the image skew must be corrected to improve recognition accuracy. Three is an effective algorithm to recognize image texture feature must be constructed.

This paper focuses on these problems and proposes a new computer recognition algorithm for image texture, and made recognition result higher accuracy with less calculation and extensive application. Firstly, 2-d bidirectional median filtering is proposed to make image de-noising an ideal state. Secondly, an image correction algorithm with Hough transfer on local calculation is presented. The image reaches a better position state. A method based on local extreme value is proposed to search for image texture feature and recognize image texture parameter. Finally, this method is validated through fabric density recognition, and has a better recognition effect.

## II. 2-D BIDIRECTIONAL MEDIAN FILTERING

The clearer image texture is, the more accurate the subsequent analysis is. So the image de-noising is very important. The median filtering has better restraining effect in disturbing impulse and point noise. However, the median filtering is very flexible, and the value of its window length and dimension are various. In this study, combing image characteristic, we propose a novel 2-d bidirectional median filtering approach, which can made image noise elimination.

As shown in Figure1, the point  $P$  is the point to be treated. Let lateral window length is  $L_w$ , longitudinal

window length is  $L_k$ ,  $v_w$  be the number of lateral window pixel value,  $v_k$  be the number of longitudinal window pixel value.

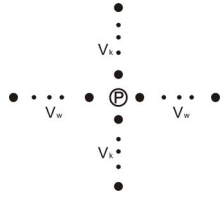


Fig.1. Schematic diagram of 2-d bidirectional median filtering

Let  $v_w = (L_w - 1)/2$ ,  $v_k = (L_k - 1)/2$ , the sequence of lateral window pixel point according to the gray levels from small to big is  $\{w_{i-v_w}, \dots, w_{i-1}, w_i, w_{i+1}, \dots, w_{i+v_w}\}$ , the sequence of longitudinal window pixel point according to the gray levels from small to big is  $\{k_{j-v_k}, \dots, k_{j-1}, k_j, k_{j+1}, \dots, k_{j+v_k}\}$ . Let the output value of lateral median filtering is  $G_i$ , the output value of longitudinal median filtering is  $G_j$ , they can be expressed as:

$$G_i = \text{Med}\{w_{i-v_w}, \dots, w_{i-1}, w_i, w_{i+1}, \dots, w_{i+v_w}\} \quad (1)$$

$$G_j = \text{Med}\{k_{j-v_k}, \dots, k_{j-1}, k_j, k_{j+1}, \dots, k_{j+v_k}\} \quad (2)$$

Where,  $\text{Med}$  denotes the middle value of the sequence. Let the output of the 2-d median filtering is  $G(i, j)$  as shown in equation 3:

$$G(i, j) = G_i \times \frac{L_w}{L_w + L_k} + G_j \times \frac{L_k}{L_w + L_k} \quad (3)$$

### III. HOUGH TRANSFORM ON LOCAL COMPUTATION

Even though the image inclination angle is only  $0.1^\circ$ , it is obvious detected by human eyes and can generate huge effect to texture recognition result. Therefore, it is necessary to correct the image inclination. Hough transform is an effective line detection method, which is less influenced by image noise and discontinuous, and is appropriate to image texture treatment. But its features are high accuracy and large calculation. In order to reduce the calculation, Le[18], wag[19] proposed different algorithms, but the result is not ideal. In this study, we find that image inclination has accordant characteristic in local and whole image. Thus, the local analysis method is used to reduce calculation. That is, the inclination angle is detected only selecting a smaller region in an image. So this inclination angle denotes the one of entire image.

Let the random point of image space be  $(x_0, y_0)$ , the formula of Hough transform is given as:

$$\rho = x \cos \theta + y \sin \theta \quad (4)$$

From the formula (4), the point  $(x_0, y_0)$  can transform to a sinusoid in the parameter space  $(\rho, \theta)$ , and the sinusoid is  $\rho = x_0 \cos \theta + y_0 \sin \theta$ . The corresponding sinusoid between random two points in the image space

can intersect to a point  $(\rho_0, \theta_0)$  in the parameter space. If a line  $l$  connected these two points exist, then the sinusoid of each point on line after Hough transform can intersect to point  $(\rho_0, \theta_0)$ . Therefore, the point  $(\rho_0, \theta_0)$  is found that the line  $l$  is found. The problem of line detection in image space is transformed to searching for the local maximum value in parameter space. The detail work is to establish the accumulator matrix in the parameter space, and the parameter curve of each point in the image space is calculated. Then the matrix value of all points on the parameter curve is added 1.

If the line  $l$  is existed in the image, a corresponding element existing in the accumulator matrix is the local maximum. The position  $(\rho_0, \theta_0)$  of the maximum is found, the corresponding line  $l$  in the image space is obtained.  $\rho_0$  and  $\theta_0$  are the intercept and slope of the line.

From above principle, if there have a set of parallel lines, the sinusoid of the point on the parallel lines will frequently intersect at a point. A set of local maximum will be existed around the same inclination angle in the parameter space. Considering image has regulation characteristic, the corresponding point's  $(\rho_{\max}, \theta_{\max})$  angle  $\theta_{\max}$  of maximum calculated by accumulator is the slope of the parallel lines.

In order to reduce the computing speed, the slope of line is detected by selecting one region of image. The region is prescribed that the width is 10cm, the high is 10cm. The pixel number is calculated by the solution, then Hough transform is made and construct the accumulator matrix to search for the slope  $\theta_{\max}$  of the parallel beelines. Finally, the image is corrected.

### IV. TEXTURE MATRIX AND POSITION MATRIX ON LOCAL EXTREME VALUE

This paper constructs image texture matrix and position matrix according to the gray local extreme value in horizontal and vertical direction. The texture parameters needed are obtained according to some corresponding formulas from two matrixes. The gray of any texture image watched by naked eyes has regular change in horizontal and vertical direction. Figure1 shows an image texture style with uniform change. Up and down ellipses are local gray maximum and minimum respectively. These local extreme values determine the image texture feature. Also the local extreme value determines the image texture feature for the image with complicated rule change.

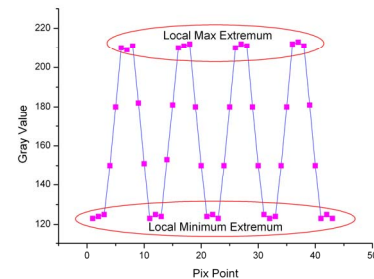


Fig.1 Local extreme value of image texture

Let pixel points appear continuously in lateral direction be  $p_1, p_2, \dots, p_m$ , its gray values are  $g_1, g_2, \dots, g_m$ , the gray local maximum and local minimum is easily obtained by cycle judging the gray value around each point with computer program. Considering the image local gray extreme value may be abnormal or catastrophe points occur, two points around each extreme value obtained in first step are selected again, and the gray mean value of five points is obtained. comparing the mean value with two continuous points except the left and right two points, the mean value still is local extreme value and the correctness of this extreme value is affirm. In the same reason, the gray local extreme value in vertical direction is obtained according to above methods.

All local extreme values in weft and warp direction are made up of fabric texture feature matrix denoted by gray. let fabric texture feature matrix be  $T_t$ ,  $N$  be the row number of matrix,  $M$  be the array number of matrix,  $g(x_i, y_j)$  denotes the gray on the  $i$ th row and the  $j$ th array. Matrix  $T_t$  is

$$T_t = |g(x_i, y_j)|_{N \times M} \quad (5)$$

Let  $T_s$  denotes the position of each local extreme value, the fabric image texture position matrix is

$$T_s = |p(x_i, y_j)|_{N \times M} \quad (6)$$

## V. EXAMPLE AND DISCUSSION

Fabric density analysis is a typical image texture recognition example. For long this work is commonly made by manual work with simple implement, it not only wastes time but also easy to make mistakes. In this paper, we obtain a better effect with an image texture recognition algorithm on median filtering and Hough transform. The system is programmed by Matlab7.0 according to the algorithm proposed in this paper. The fabric image is captured by the high definition camera and recognized automatically density by computer.

### A. 2-d Bidirectional Median Filtering Treatment

As is shown in Figure 2, there are four samples: Sample A, Sample B, Sample C, and Sample D. They are treated by 2-d bidirectional median filtering. Using this method, the fabric image is enhanced for image de-noising. The graphs which name with “-1” are the former graphs, the graphs which name with “-2” are result recognition graphs using 2-d bidirectional median filtering treatment. After comparison, we find the image texture become clearer by 2-d bidirectional median filtering.

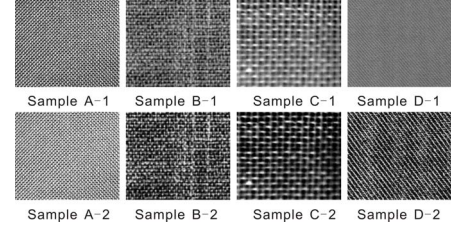


Fig.2 Recognition result using 2-d bidirectional median filtering

### B. Local Hough Transform Treatment

Also using above four samples, the images are treated with Hough transform as shown in Figure 3. The graphs which name with “-2” are the former graphs, the graphs which name with “-3” are result recognition graphs using Hough transform treatment. Obviously, the image texture itself has not distinct incline, such as sample B and sample D, the texture inclination is similar after Hough transform treatment. However, the image texture has distinct incline, such as sample A and sample C, the texture inclination is corrected by Hough transform treatment. It shows that the local Hough transform can judge automatically the fabric image inclination. When the image has incline, it can give adjustment. When the image has not any incline, it can keep the original state.

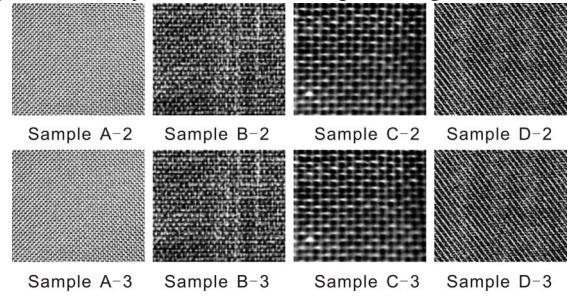


Fig.3 Recognition result using Hough transform

### C. Matrix Construction on Local Extreme Value and Recognition Result

A clearer image is obtained after median filtering and Hough transform treatment. The texture matrix and position matrix indicating texture feature are constructed using the local extreme value searching proposed in this paper. From the extreme value position matrix, fabric weft and warp density can obtained easily using formulas as follows:

$$D_{weft} = \frac{M}{S_M} \times 10 \quad (7)$$

Where  $D_{weft}$  denotes weft density,  $M$  be the number of columns in local extreme value position matrix,  $S_M$  denotes corresponding image width in this position matrix (unit is cm).

$$D_{warp} = \frac{N}{S_N} \times 10 \quad (8)$$

Where  $D_{warp}$  denotes warp density,  $N$  be the number of rows in local extreme value position matrix,

$s_N$  denotes corresponding image width in this position matrix (unit is cm).

The density recognition results are shown in Table1, and the recognition comparison between computer and manual is given. From Table1, the fabric density recognition result is exact using the texture recognition algorithm proposed in this paper. The computer recognition results of Sample A and Sample C are completely similar to manual recognition. The reason is two collecting images quality are high and the structures gray are uniform. The computer recognition results of Sample B are a little different from manual recognition. The reason is the right gray distribution in the image is different from the left gray distribution. In sample D, the weft density recognition result is better, but the warp density recognition result has error. The reason is the gray in yarns interspaces is not uniform and cause warp recognition error though the structure texture is clear.

TABLE1 COMPUTER AND MANUAL RECOGNITION COMPARISONS

| Recognition method | Computer recognition |                     | Manual recognition  |                     |
|--------------------|----------------------|---------------------|---------------------|---------------------|
|                    | Weft Density (N/cm)  | Warp Density (N/cm) | Weft Density (N/cm) | Warp Density (N/cm) |
| Sample A           | 189                  | 221                 | 186                 | 219                 |
| Sample B           | 123                  | 135                 | 115                 | 121                 |
| Sample C           | 79                   | 95                  | 79                  | 96                  |
| Sample D           | 192                  | 169                 | 190                 | 143                 |

#### D. Further Discussion

When fabric density is recognized from examples, fabric density is calculated and obtained only using two parameters from local extreme value position matrix. In fact, texture matrix and position matrix constructed in this paper contain many image texture information owing to these two matrixes are constructed according to local extreme value (namely key characteristic points). Some formulas from these two matrixes can describe many texture image indexes, such as texture uniform index, texture shape and pattern change, texture quality index, texture entropy, texture roughness and so on. Thus, the algorithm proposed in this paper has extensive application to texture recognition. Here is not discussed deeply due to the limited space.

#### VI. CONCLUSION

In this paper, 2-d bidirectional median filtering is proposed to effectively eliminate texture image noise, and make image clearer for analysis. Texture image is analyzed to correct the image incline using Hough transform, so that not only large calculating is avoided but also the guarantee of texture analysis accuracy is provided. The texture image local extreme value matrix and position matrix are constructed, so that the texture feature is extracted effectively and large calculating is reduced. After proved by examples, these algorithms proposed in this paper can make density analysis quickly,

accurately and have extensive application.

#### REFERENCES

- [1] Wang Z, Bovik Alan C and Sheikh Hamid R, et al, "Image quality assessment: from error visibility to structural similarity", IEEE Transactions on Image Processing, vol.13, Apr.2004, pp.600-612, doi:10.1109/TIP.2003.819861
- [2]Tea Jin Kang,Chang Hoon Kim., "Automatic Recognition of Fabric Weave Patterns By Digital Image Analysis", Textile Research Journal, vol.69, Feb.1999, pp.77-83,doi: 10.1177/004051759906900201
- [3] D.R. Rohrmus, "Invariant and adaptive geometrical texture features for defect detection and classification", Pattern Recognition , vol.38, Feb.2003, pp.1546-1599, doi: doi:10.1016/j.patcog.2005.02.004
- [4] Radovan Stojanovicw, Panagiotis Mitropulos,et al, "Real-Time Vision-Based System for Textile Fabric Inspection", Real-Time Imaging, vol.7, Dec.2001,pp.507-518, DOI: 10.1006/rtim.2001.0231
- [5] Chung-Feng Jeffrey Kuo,Te-Li Su and Yi-Jen Huang, "Computerized color separation system for printed fabrics by using backward-propagation neural network[J], Fibers and Polymers,2007,8(5):529-536,doi: 10.1007/BF02875876
- [6] H.K.Kwag., S.H.Kim., "Efficient skew estimation and correction algorithm for document image", Image and Vision Computing, vol.20, Jan.2002, pp.25-35, doi: 10.1016/S0262-8856(01)00071-3
- [7] Zhang J.S. and Leung Y.W., "Improved possibilistic c-means clustering algorithms", IEEE Trans. on Fuzzy Systems, vol. 12, April. 2004, pp.209-217, doi: 10.1109/TFUZZ.2004.825079
- [8] Josso B, Burton D R and Lalor M J., "Texture Orientation and Anisotropy Calculation by Fourier Transform and Principal Component Analysis", Mechanical Systems and Signal Processing, vol.19, May.2005, pp.1152-1161, doi:10.1016/j.ymssp.2004.07.005
- [9] Chan Chi-ho, "Defect Detection by Fourier Analysis", IEEE Transactions on Industry Applications, vol.36, Oct.2000, pp.1267-1276, doi:10.1109/28.871274
- [10]Arivazhagan S and Ganesan L, "Texture Classification Using Wavelet Transform", Pattern Recognition Letters, vol.24, Jun.2003, pp.1513-1521, DOI: 10.1016/S0167-8655(02)00390-2
- [11] Soo Chang Kim and Tae Jin Kang, "Texture classification and segmentation using incomplete tree structured wavelet packet frame and Gaussian mixture model", Vol. 13, May.2005,pp. 46- 51, doi:10.1109/IST.2005.1594525
- [12]SHEN Jianqiang,GENG Zhaofeng and ZOU Xuan, "Detection Method for Fabric Texture Direction Based on Wavelet Transform", Computer Engineering, vol.33, Jun.2007, pp.182-184,doi: CNKI:ISSN:1000-3428.0.2007-06-063
- [13]Haihong Wu, Mingmin Zhang and Zhigeng Pan, "Automatic Identifying Weave Patterns for Double-layer Weft Woven Fabric", Proceedings of the International Conference on Computer Graphics, Imaging and Visualization, July.2005, pp.117-122, doi:10.1109/CGIV.2005.24
- [14]Bodnarova, Bennamoun and S. Latham, "Optimal Gabor !lters for textile Saw detection", Pattern Recognition, vol.35, Dec.2002,pp.2973-2991, DOI: 10.1016/S0031-3203(02)00017-1
- [15]K.L. Mak and P. Peng, "An automated inspection system for textile fabrics based on Gabor filters", Robotics and Computer-Integrated Manufacturing ,vol.24, Jun.2008, pp.359-369, doi: 10.1016/j.rcim.2007.02.019
- [16] Ong E P. "Robust optical flow computation based on least median of squares regression", Computer Vision, vol.31,Jan.2003, pp.51-82
- [17]LI Jia-yan,DING Yong-sheng and ZHANG Zhong-wei, "Fabric texture image identification method for CCD-based weft-straightening machine", Computer Engineering and Applications, vl.43, July.2007pp.214-216,doi: CNKI:ISSN:1002-8331.0.2007-13-062
- [18] D.X.Le,G.Thoma and H.Weschler, "Automated page orientation and skew angle detection for binary document image", Pattern Reconition, vol.27, Oct.1994, pp.1325-1344
- [19]H.K.Kwag. and S.H.Kim. "Efficient skew estimation and correction algorithm for document image", Image and Vision Computing, vol.20, Jan.2002, pp.25-35,doi: 10.1016/S0262-8856(01)00071-3